MANIPAL ACADEMY of HIGHER EDUCATION
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COMPARATIVE DEEP LEARNING APPROACHES FOR NETWORK INTRUSION DETECTION

Synopsis Report on Deep Learning Project [ICT-4442]

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August 2026

PART A : SYNOPSIS TEMPLATE

Team No. and Names (with Registration Numbers):

Team No.: To be assigned
1. **Aditi Arun** – Registration No. 230911108
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Title of the Project:

Comparative Deep Learning Approaches for Network Intrusion Detection

Brief Description of the Project:

Modern computer networks face a growing volume and diversity of cyberattacks, ranging from denial-of-service floods to brute-force login attempts and botnet-driven traffic. Traditional intrusion detection approaches, which rely on hand-crafted rules or signature matching, struggle to keep pace with attacks that constantly evolve and with the sheer scale of traffic generated on real networks. This creates a need for detection methods that can automatically learn discriminative patterns directly from high-dimensional network-flow data. Deep learning offers a promising direction for this problem, as neural networks can learn complex, non-linear relationships among traffic features without requiring the detection rules to be manually specified in advance.

This project aims to build and compare four deep learning architectures for network intrusion detection: a Multilayer Perceptron (MLP) as a feed-forward baseline, a 1D Convolutional Neural Network (1D CNN) to capture local patterns and feature interactions, a Long Short-Term Memory (LSTM) network to explore sequential/recurrent modelling of traffic records, and a Transformer/attention-based model to investigate whether self-attention can capture global relationships among input features more effectively than the other architectures. All four models will be trained and evaluated on the same dataset split under a common experimental protocol, so that differences in performance can be attributed to the architecture rather than to inconsistencies in the experimental setup.

Performance will be assessed using standard classification metrics such as accuracy, precision, recall, and F1-score, computed for both the benign and attack classes. Beyond simply reporting which model achieves the highest accuracy, this project places particular emphasis on analyzing why the architectures perform differently — for instance, whether convolutional filters are better suited to capturing localized feature interactions in flow-based data, or whether recurrent and attention mechanisms offer any advantage when traffic records are treated as sequences. This comparative and explanatory approach is intended to yield practical insights into which classes of deep learning models are best suited to network intrusion detection, a problem of direct relevance to the cybersecurity industry.

Dataset(s) Identified:

The project will use the **CIC-IDS2017 dataset** (Canadian Institute for Cybersecurity Intrusion Detection Evaluation Dataset), accessed via a cleaned/preprocessed version available on Kaggle. The dataset consists of network-flow records generated from captured network traffic, where each record is described by a set of extracted flow-level features (such as flow duration, packet counts, byte counts, and inter-arrival time statistics) together with a label indicating whether the flow is benign or represents a specific type of cyberattack. Based on the dataset documentation, attack traffic in CIC-IDS2017 is broadly associated with categories such as DoS/DDoS attacks, brute-force attacks, botnet activity, and web attacks, though the exact set of attack classes, feature columns, and class balance present in the specific Kaggle version used will be confirmed during dataset acquisition and exploratory data analysis. The final subset of data, the classes retained for classification, and the preprocessing configuration (e.g., feature scaling, handling of class imbalance, and the train/validation/test split) will be finalized at that stage and reported in detail in the Interim submission.

Proposed Models for Comparison:

1. **Multilayer Perceptron (MLP):** A feed-forward neural network used as the baseline model for the comparison.
2. **1D Convolutional Neural Network (1D CNN):** Used to learn local patterns and feature interactions within the network-flow feature vectors.
3. **Long Short-Term Memory (LSTM) network:** The recurrent architecture used to explore sequential dependencies in the traffic data.
4. **Transformer / Attention-based model:** Used to investigate self-attention and global relationships among the input features.

Model-to-Member Assignment:

- **Aditi Arun:** 1D CNN (implementation, training, and evaluation).
- **Anushka Thakur:** LSTM (implementation, training, and evaluation).
- **Joint Assignment:** MLP baseline and Transformer/attention-based model are jointly handled by both team members, given the team currently has two members against four proposed models; final individual ownership of these two models will be documented in the Interim contribution log and reflected in the GitHub commit history.

Existing State-of-the-Art Method(s) Related to the Proposed Project:

Recent research on network intrusion detection has explored a range of deep learning approaches. These include deep feed-forward neural networks trained on extracted flow features, convolutional neural network (CNN)-based detectors that treat traffic features as structured input to capture local patterns, and recurrent neural networks/LSTM-based models that treat traffic as sequential data. Several works have also proposed hybrid CNN-LSTM architectures that combine local feature extraction with sequential modelling, as well as more recent attention-based and Transformer-based approaches that aim to capture global dependencies among features. No single architecture family is treated here as the definitive state of the art; rather, this body of literature indicates that different architectures make different modelling assumptions about network traffic. This project builds on that observation by comparing representative architectures from four distinct deep learning families — MLP, 1D CNN, LSTM, and Transformer — under a single, common experimental protocol on the same dataset, which allows their relative strengths and weaknesses for intrusion detection to be assessed directly.

Objectives:

1. To evaluate and compare different deep learning architecture families (MLP, 1D CNN, LSTM, and Transformer) for network intrusion detection using a common dataset and evaluation protocol.
2. To identify the model that provides the best overall intrusion-detection performance based on appropriate metrics such as accuracy, precision, recall, and F1-score.
3. To analyze how differences in neural network architecture influence performance across benign and malicious network traffic classes.

Importance / Motivation of the Proposed Project:

Cybersecurity has become increasingly critical as the scale and sophistication of network attacks continue to grow, and the volume of traffic on modern networks makes manual or purely rule-based monitoring difficult to sustain. Deep learning offers the potential to automatically learn complex, high-dimensional patterns in network traffic that may be difficult to capture with handcrafted rules. By comparing multiple deep learning architectures under identical experimental conditions, this project aims to generate practical insight into which types of models are better suited to intrusion-detection tasks, rather than assuming that any single architecture is universally superior. This comparative approach to applied deep learning is directly relevant to the AI, cybersecurity, cloud, and data engineering domains, and reflects the kind of applied model-selection and evaluation skills expected in technical software and security roles in industry.

Preliminary Reference List:

[1] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," in *Proc. 4th Int. Conf. Information Systems Security and Privacy (ICISSP)*, 2018, pp. 108–116.

[2] N. Shone, T. N. Ngoc, V. D. Phai, and Q. Shi, "A Deep Learning Approach to Network Intrusion Detection," *IEEE Trans. Emerging Topics in Computational Intelligence*, vol. 2, no. 1, pp. 41–50, 2018.

[3] R. Vinayakumar, M. Alazab, K. P. Soman, P. Poornachandran, A. Al-Nemrat, and S. Venkatraman, "Deep Learning Approach for Intelligent Intrusion Detection System," *IEEE Access*, vol. 7, pp. 41525–41550, 2019.

[4] M. Yin, Y. Chong, and W. Fu, "A Deep Learning Approach for Intrusion Detection Using Recurrent Neural Networks," *IEEE Access*, vol. 5, pp. 21954–21961, 2017.

[5] Y. Yang, K. Zheng, C. Wu, X. Niu, and Y. Yang, "Building an Effective Intrusion Detection System Using the Modified Density Peak Clustering Algorithm and Deep Belief Networks," *Applied Sciences*, vol. 9, no. 2, art. 238, 2019.

[6] P. Wu and H. Guo, "LuNET: A Deep Neural Network for Network Intrusion Detection," in *Proc. IEEE Symp. Series on Computational Intelligence (SSCI)*, 2019, pp. 617–624.

Work Plan:

Sl. No.	Component / Work Element / Milestone	Expected Completion
1	Project topic finalization, dataset selection, and GitHub repository setup	26/08/2026
2	Dataset acquisition, exploratory data analysis, cleaning, preprocessing, encoding, normalization, and train/validation/test split	05/09/2026
3	Literature review and compilation of 8–10 relevant papers	10/09/2026
4	MLP baseline and 1D CNN implementation with preliminary evaluation	18/09/2026
5	LSTM implementation with preliminary evaluation	21/09/2026
6	Interim results compilation, contribution log, and Interim Report preparation	24/09/2026
7	Interim Report Submission	25/09/2026
8	Transformer / attention-based model implementation	05/10/2026
9	Hyperparameter tuning and standardized evaluation of all four models	13/10/2026
10	Comparative performance analysis, error analysis, and final report writing	23/10/2026
11	Presentation preparation, viva practice, and in-class Final Presentations	26–29/10/2026
12	Final Report and Code Repository Submission	31/10/2026